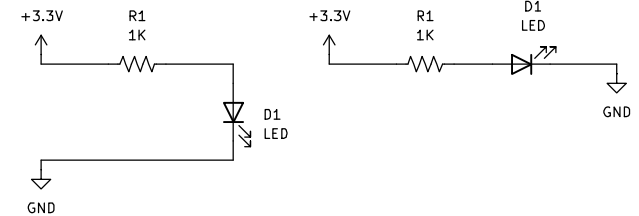
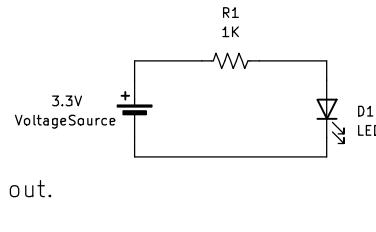
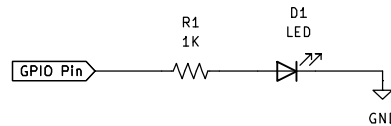


LED w/ Current-Limiting Resistor

- LEDs turn on when current flows through them.
- Resistor in series limits the current flowing through LED.
- LEDs typically have a max current rating of approx. 20mA
- CAUTION: If the maximum current is exceeded, LED will burn out.
- every LED needs it's own current limiting.
- increasing resistor value decreases current to LED, reducing brightness of LED.

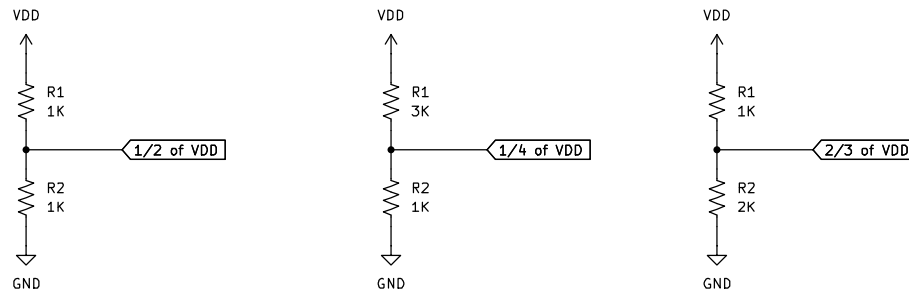


MCU-controlled LED



Turn LED ON & OFF with digital control via GPIO Pin of Micro Controller Unit (MCU).
When GPIO pin's voltage goes high, LED turns ON.

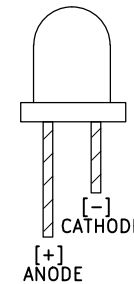
VOLTAGE DIVIDER



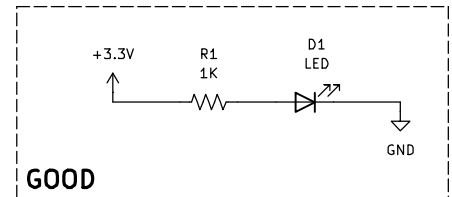
By adjusting the ratio of resistor values R1 & R2,
we can derive a new static voltage within the range of Ground (0V) and VDD.

If R1 & R2 are equal, then the output of voltage divider will be 1/2 of VDD.

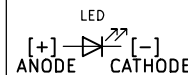
LED Polarity



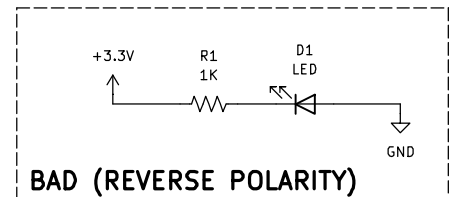
GOOD



LEDs have POLARITY.
Current will only flow in one direction, from:
positive Anode [+] to negative Cathode [-].
– Long Leg = ANODE [+]
– Short Leg = CATHODE [-]



BAD (REVERSE POLARITY)



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